5.4.3 Planetary motion

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	Kepler's three laws of planetary motion	
(b)	the centripetal force on a planet is provided by the gravitational force between it and the Sun	
(c)	the equation $T^2 = \left(\frac{4\pi^2}{GM}\right)r^3$	Learners will also be expected to derive this equation from first principles.
(d)	the relationship for Kepler's third law $T^2 \propto r^3$ applied to systems other than our solar system	
(e)	geostationary orbit; uses of geostationary satellites.	HSW1, HSW2 Predicting geostationary orbit using Newtonian laws.